

Highlights

Shoreline compactness predicts Sentinel-1 water-area monitoring reliability across reservoirs worldwide

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- Shoreline area-to-perimeter ratio (A/P) predicts SAR accuracy in advance
- Median KGE is 0.93 above 200 m A/P but only 0.48 below 100 m
- Climate adds a secondary control on accuracy beyond A/P ($p=0.012$)
- A simple per-scene VV detector matches a costlier dual-pol SVM overall
- An open GEE app flags 35,000+ reservoirs by expected reliability